SANDIP FOUNDATION

Project Stage –I: Final Presentation

On

“RansomShield AI: An AI-Powered Endpoint Threat Detection, Response, and Recovery System”

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Academic Year (2026 – 27)

Department of Computer Engineering

Sandip Institute of Technology and Research Centre, Nashik.

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Evidence-status key used in this deck:

Observed in prototype | Design component | Stage-II work package | Research basis

Sponsorship Letter

SANDIP FOUNDATION

BOSCH

Ref. No.: 30671568

Date: August 13, 2026

To,
The Head of Department,
Department of Computer Engineering,
Sandip Institute of Technology and Research Centre (SITRC),
Nashik, Maharashtra.

SUBJECT: INDUSTRIAL SPONSORSHIP AND PROJECT COLLABORATION LETTER FOR B.E. FINAL YEAR PROJECT- "AI-DRIVEN REAL-TIME RANSOMWARE INTERCEPTION AND FILE ROLLBACK SYSTEM"

Respected Sir/Madam,

We are pleased to inform you that BOSCH Limited, *Nashik Plant* has reviewed and approved the proposed industrial collaboration and project sponsorship for the final-year engineering project titled "AI-Driven Real-Time Ransomware Interception and File Rollback System."

The project is being undertaken by the following final-year students from the Department of Computer Engineering:

Sr. No.	Student Name	Project Role
1	Lakansing Narsing Patil	Backend & Microservices Architect
2	Rutuja Pankaj Patil	User Interface & System Analysis
3	Srushti Vinayak Khapre	Testing / Security Devloper
4	Vaishnavi Sunil Thapekar	AI & Module Integration

The proposed system focuses on developing a cybersecurity solution capable of detecting ransomware-like activity in real time, intercepting suspicious file operations, and enabling recovery or rollback of affected files. The project aims to explore the practical application of cybersecurity, machine learning/AI techniques, file-system monitoring, threat detection, and automated response mechanisms.

As part of the proposed industrial collaboration, BOSCH may provide appropriate technical guidance, subject-matter inputs, infrastructure support, or project-related mentoring, subject to its internal policies and approval procedures.

The collaboration is intended to provide the students with practical exposure to industry-oriented cybersecurity practices, secure software development, threat detection, system design, testing, and incident-response concepts.

We appreciate the efforts of the students and their academic department in undertaking this industry-relevant engineering project and wish them success in their technical work and future professional journey.

Yours faithfully,

pki, BOSCH, APAC, S, Sneha.Oak — Digitally signed by pki, BOSCH, APAC, S, N, S=Sneha.Oak

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Project Background & Motivation

- **Literature:** ransomware locks systems or data and demands payment; crypto-ransomware encrypts documents, photos, databases and archives [1], [2].
- **Literature:** signature-based antivirus may not react quickly to zero-day, polymorphic, obfuscated or fileless variants, so many files can change before detection (paper §I).
- **Literature:** behaviour-based detection watches what a process does, but backup, compression, installer and sync tools also create heavy file activity [4]–[6]; file entropy is a complementary signal [7].
- **Team objective:** combine detection, response, recovery support and central visibility in one prototype, RansomShield AI. This is a goal, not a literature finding.

1 Detect	2 Respond	3 Recover	4 Observe
Behavioural + entropy monitoring of process and file activity	Confidence-based decision; staged interception	Protected versions, selective rollback, hash check (proposed)	Sync to cloud; dashboard for administrator / SOC analyst

Problem Definition & Need

Research question (from the paper)

Can a lightweight hybrid model detect ransomware early enough to minimise file damage while keeping false positives low during legitimate file-intensive workloads?

Target environment

Windows endpoints and small organisational environments (paper abstract). Prototype test machine: Windows 11 (Device Fleet page).

Why existing approaches fall short (paper §I–II)

- Static or file-based patterns can be invalidated by obfuscation, packing and polymorphism.
- Behavioural detectors overlap with legitimate software [4]–[6]; high entropy alone is insufficient because archives and legitimately encrypted files also score high [7].
- Classification accuracy alone does not show whether detection happens before significant damage.

Technical needs (project)

1. Early detection from short-window features
2. Low false alarms on benign heavy file work
3. Link activity to the responsible process
4. Restore only affected files, hash-verified
5. Central, authenticated visibility

Existing System / Current Solutions

File-based / signature

Inspects executable structure or known content patterns. Effective for previously analysed families; obfuscation, packing, polymorphism and metamorphism can invalidate static patterns (paper §II-A).

Network-based

Monitors command-and-control traffic. Encrypted standard protocols, anonymous services and legitimate cloud platforms can hide malicious traffic; SDN-based mitigation is studied in [25].

Behaviour-based

Observes API calls, registry changes, process creation, rapid file access, extension changes and repeated writes [4]–[6]. Modifying user files is hard to hide; the limitation is overlap with legitimate software.

Interception & recovery defences

CryptoDrop [9] detects suspicious file transformations to stop ransomware early; ShieldFS [10] adds a self-healing file system; RWGuard [28] combines process and file-system monitoring with decoys.

Positioning: RansomShield AI is a student-built prototype of the paper's proposed framework. Categories above summarise the paper's related work; no superiority over commercial antivirus or EDR products is claimed.

Existing System / Current Solutions

Approach	Technique	Strength	Limitation (as stated in the paper)
File-based / signature	Executable structure, known content patterns	Effective for previously analysed families	Obfuscation, packing, polymorphism invalidate static patterns
Network-based [25]	C&C traffic monitoring; SDN flow control	Timely mitigation by controlling flows	Encrypted protocols, anonymisation and cloud platforms hide traffic
Behaviour-based [4]–[6]	API calls, registry, process creation, rapid file access	File modification cannot be completely hidden	Behavioural overlap with legitimate software
ML-based [7], [8], [27]	Entropy or dynamic features with SVM, trees, forests	Learns patterns beyond fixed rules	Frameworks, datasets and operational concerns need comparison [8]; entropy alone insufficient [7]
Interception + recovery [9], [10], [28]	File-transformation detection, self-healing file system, decoys	Rapid interception and recovery	Paper's stated gap: an implementable end-to-end pipeline

Note: summarised from the paper's related-work section; direct superiority claims over commercial products are not made.

Literature Review – Key Findings

Literature finding	Influence on the proposed framework (status)
[7] Entropy, compression and bit-level features improved SVM detection of encrypted files, including partial encryption.	Entropy / content feature group (Shannon, Rényi, compression ratio) – proposed.
[4]–[6] Behaviour-based detection overlaps with legitimate software.	Confidence-based risk engine with consecutive-window confirmation – proposed.
[9], [10] Rapid interception and self-healing recovery can limit damage.	Staged interception and attack-window selective rollback – proposed; one logged restore observed.
[8] Reviews DT, RF, SVM and others; calls for comparison of frameworks and datasets.	Compare LR, DT, RF, SVM and behaviour-only vs entropy-only vs hybrid – planned.
[1]–[3] NIST guidance separates identify/protect, detect/respond and recover.	Detection, containment and recovery treated as one pipeline; cloud dashboard is a project extension.

Source categories: the manuscript lists 30 references. Eleven are research publications ([4]–[10], [25]–[28]); the others are NIST standards and guidance, vendor threat reports, framework documentation and tool documentation.

Literature Review

Ref.	Authors, year (venue)	Method / focus	Key finding / limitation (as described in the paper)
[4]–[6]	Oz; Vehabovic; Ozer, 2020–23 (arXiv)	Ransomware surveys; prevention/detection system	Support behaviour-based detection; limitation is overlap with legitimate software
[7]	Hsu et al., 2021 (IEEE Access)	Entropy, compression, bit features; polynomial SVM	22 file formats, four families; > 92 % detection individually, ≥ 85.17 % combined (published results)
[8]	Smith et al., 2022 (IEEE Access)	Review of ML for ransomware detection	Covers DT, RF, SVM, recurrent models; calls for more comparison of frameworks and datasets
[9]	Scaife et al., 2016 (ICDCS) – CryptoDrop	Detects suspicious file transformations	Attempts to stop ransomware before extensive damage
[10]	Continella et al., 2016 – ShieldFS	Detection + self-healing file system	Can recover affected data
[26]	Kharraz et al., 2015 (DIMVA)	Long-term empirical behaviour study	Motivates the feature design
[27]	Sgandurra et al., 2016 (arXiv) – EldeRan	ML on early dynamic behaviour	Host-based ML detection
[28]	Mehnaz et al., 2018 (RAID) – RWGuard	Process + file-system monitoring, decoys	Real-time detection of cryptographic ransomware

Entries follow the manuscript's reference list; see slide 38.

Comparative Analysis of Literature

Study	Algorithm	Published result	Evaluation context (as reported)	RansomShield AI
ElderRan [27], via [8]	Regularised logistic regression	93.3 % average detection rate	Top 100 dynamic features; family-oriented	**Stage-II evaluation**
RANDS, via [8]	Decision tree + naive Bayes	96.27 % accuracy; 1.32 % mistake rate	Real-time assessment of zero-day variants	**Stage-II evaluation**
Cusack et al., via [7]	Random forest	86.0 % detection rate	Binary classifier on network-traffic features	**Stage-II evaluation**
Hsu et al. [7]	Polynomial-kernel SVM	> 92 % per family; ≥ 85.17 % combined	17 entropy/content features, 22 formats, 4 families	**Stage-II evaluation**
RansomWall, via [8]	Gradient tree boosting	98.25 % detection; near-zero false positives	Family-held-out, 12-fold, cryptographic families	**Stage-II evaluation**

Caution: these are literature benchmarks from different datasets, features, families and validation methods, so they are not directly comparable and are not results of RansomShield AI.

Research Gap & Final Objectives

Research / engineering gaps

G1 No single implementable pipeline links monitoring, hybrid features, process-level interception, protected rollback, alerts and controlled learning (paper §II-D).

G2 Accuracy alone does not show detection before damage; time to detection, files changed, files restored, false alarms per hour and overhead are needed.

G3 False positives during legitimate file-intensive work; entropy alone is insufficient [4]–[7].

G4 Process attribution and selective, verified recovery inside an attack window [9], [10].

G5 Project need (not from the paper): analyst-readable evidence and multi-device ownership.

Objectives (paper contributions C1–C5)

O1 (paper C1) Host-based architecture combining behavioural telemetry and file-entropy analysis.

O2 (C2) Short time-window features for early detection.

O3 (C3) Confidence-based risk engine linking behaviour to the responsible process.

O4 (C4) Attack-window-aware selective rollback with hash verification.

O5 (C5) Evaluation protocol: classification, latency, damage, recovery, overhead.

O6 (project) Secure device enrollment, sync and dashboard.

Reviewer feedback: comments on the manuscript (experimental results, dataset detail, quantitative comparison, novelty) are addressed through the Stage-II work packages on slide 36.

Research Gap Identification

Proposed gap	Evidence / source	Status
G1 No integrated, implementable pipeline	Paper §II-D: strong results for single algorithms or file-system defences, but no end-to-end pipeline for this project.	**Research basis**
G2 Accuracy ≠ early detection	Paper §II-D: operational outcomes (time to detection, files changed before containment, files restored, false alarms/h, overhead) must be evaluated.	**Research basis**
G3 False positives on legitimate work	Behavioural overlap [4]–[6]; high entropy in archives and legitimately encrypted files [7].	**Research basis**
G4 Attribution and verified recovery	Selective rollback depends on correct PID-to-file attribution and the attack window (paper §III-D, §V); related defences [9], [10].	**Research basis**
G5 Evidence and device ownership	Project requirement; Row-Level Security per Supabase documentation [S1].	**Design component**

Note: *G1–G4 come from the paper; G5 is a project engineering need. Novelty relative to [9], [10] and [28] is discussed on slide 32.*

Gap–Objective–Proposed Solution Mapping

Gap	Objective	Proposed method (paper)	Prototype status (evidence)	Validation metric
G1	O1, O6	Eight-stage host pipeline (Fig. 1)	**Agent, backend and dashboard running**	End-to-end run success; sync success rate
G2	O5	Operational metrics: Ld, files modified before suspension, false alarms/h	**Scores displayed in the UI**	Ld, files damaged, false alarms per hour
G3	O2, O3	Behaviour + entropy features; k-window confirmation	**Feature vectors and risk score displayed**	Precision, recall, F1, PR-AUC; ablation study
G4	O4	Attack window [ts, td]; hash-verified restore	**One restore record logged**	RSR (eq. 5) and Lr (hash-verified)
G5	O6	Enrollment tokens, ownership, RLS (project design)	**Design component; 2 devices listed**	Cross-owner access denial tests

Logic: Literature → Gap → Objective → Proposed method → Prototype evidence → Validation metric

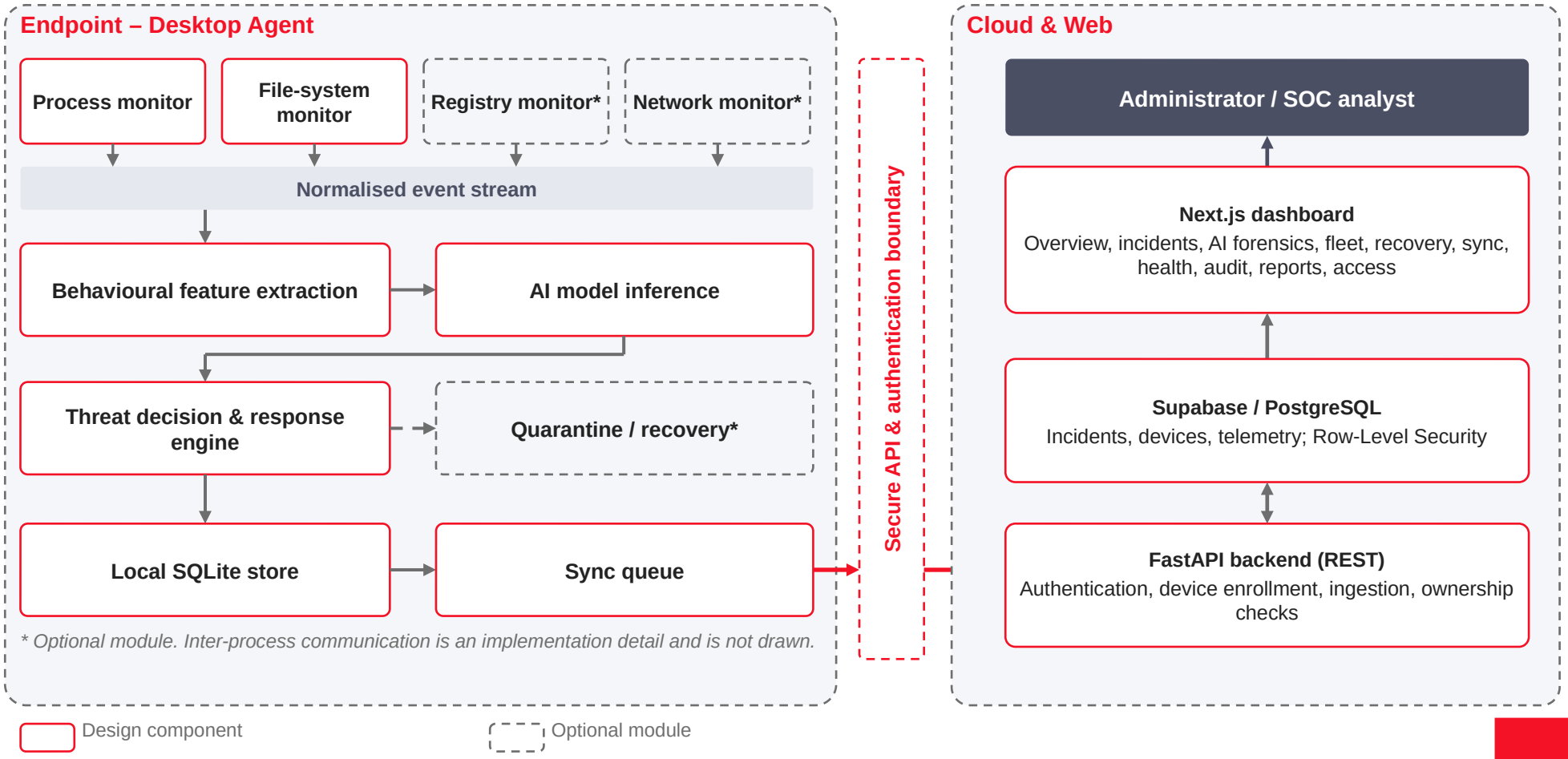
Finalized Methodology

1 Monitor	2 Collect	3 Features	4 Risk	5 Intercept	6 Window	7 Rollback	8 Report
File and process events **Observed**	PID, path, time, bytes, resources **Design scope**	Behaviour + entropy per window **Observed**	ML probability + risk score **Observed**	Suspend, then terminate **Design scope**	Attribute files in [ts, td] **Design scope**	Restore and verify hashes **Restore record**	Alert, log, incident report **Observed**

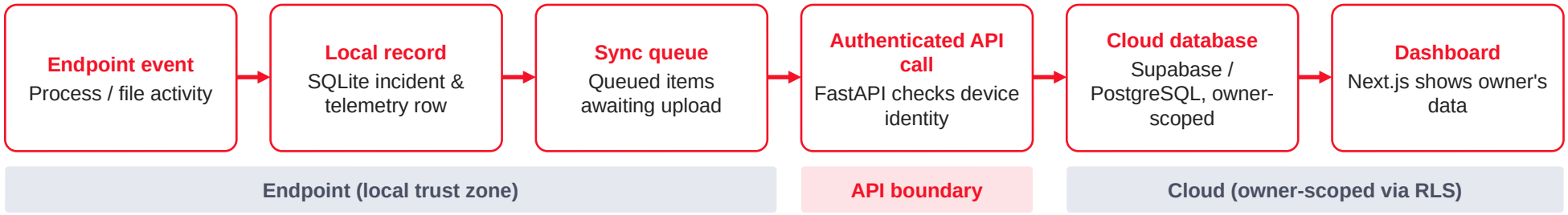
Reading this workflow

- **Source:** stages 1–8 follow the paper's proposed framework (Fig. 1). Status labels show what the prototype screenshots demonstrate.
- **Staged response:** below τs continue; between τs and τc increase sampling and snapshots; at or above τc for k consecutive windows suspend, then terminate (paper §III-B/C). Values of τs, τc and k are not given in the paper.
- **Project extension:** local SQLite queue → FastAPI → Supabase → Next.js dashboard is beyond the paper's lightweight dashboard.
- **Expert input:** no external expert validation is claimed at Stage-I (slide 31).

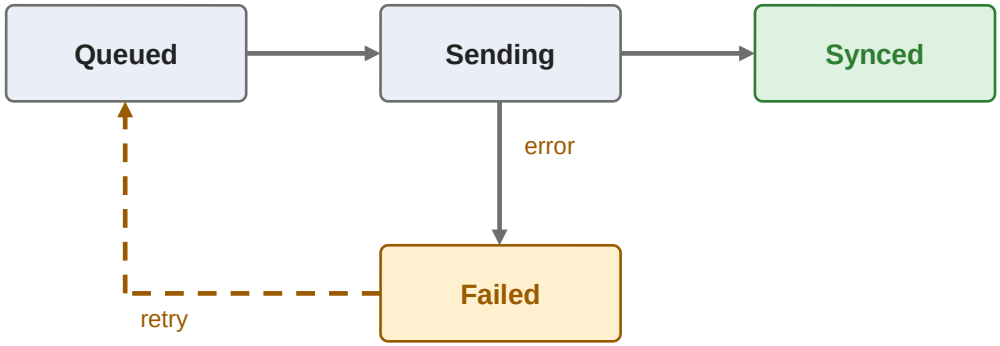
System Architecture / Block Diagram



End-to-End Data Flow & Synchronisation



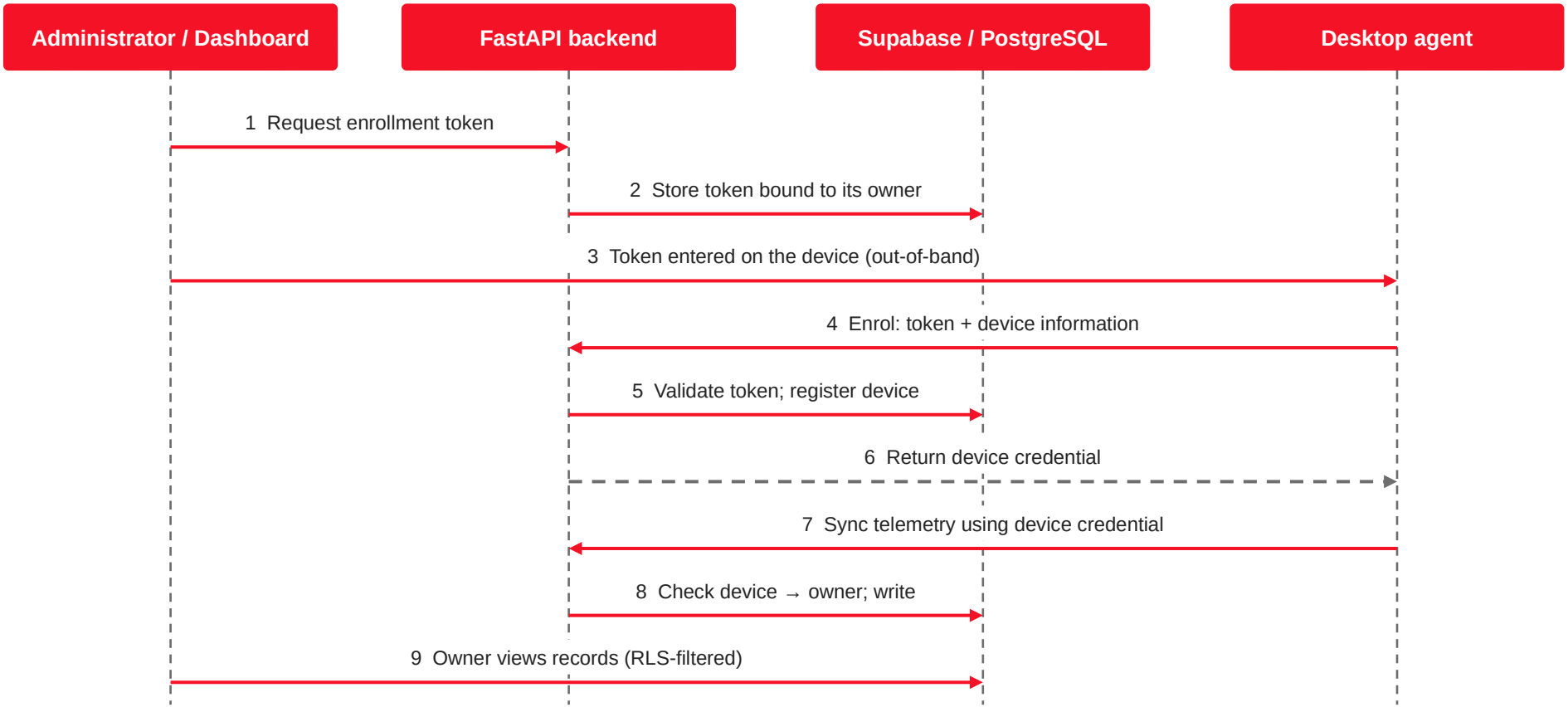
Sync states (illustrative)



Boundaries & controls (design)

- **Identity:** each record belongs to an enrolled device and its owner.
- **Authorisation:** backend checks device → owner before writing; RLS limits reads to the owner.
- **Errors:** failed uploads remain queued for retry (design intent).
- **Not claimed:** end-to-end encryption, tamper-proofing, guaranteed real-time delivery, production certification.

Device Enrollment & Ownership Design

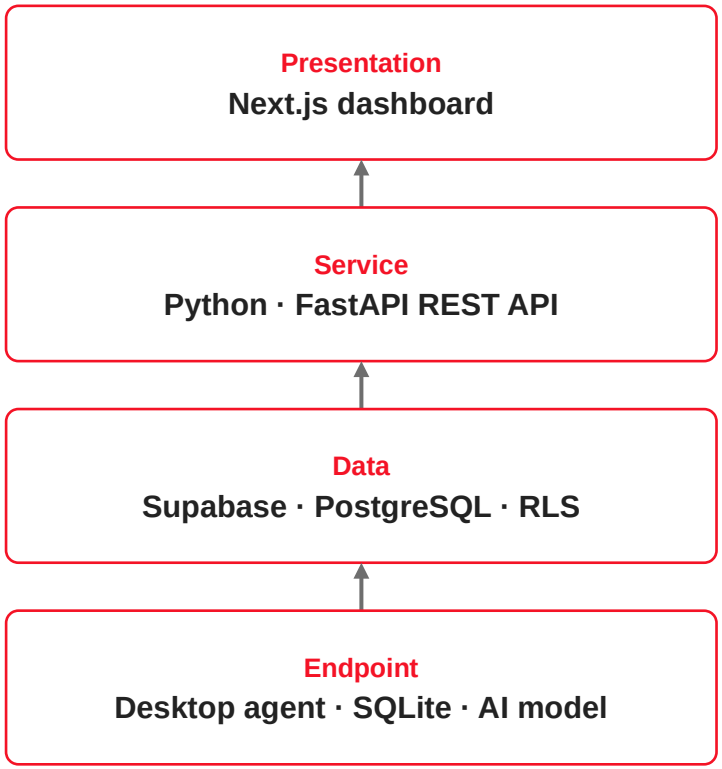


Token lifetime and credential format are implementation details and are not specified on this slide.

Design component

Hardware & Software Design

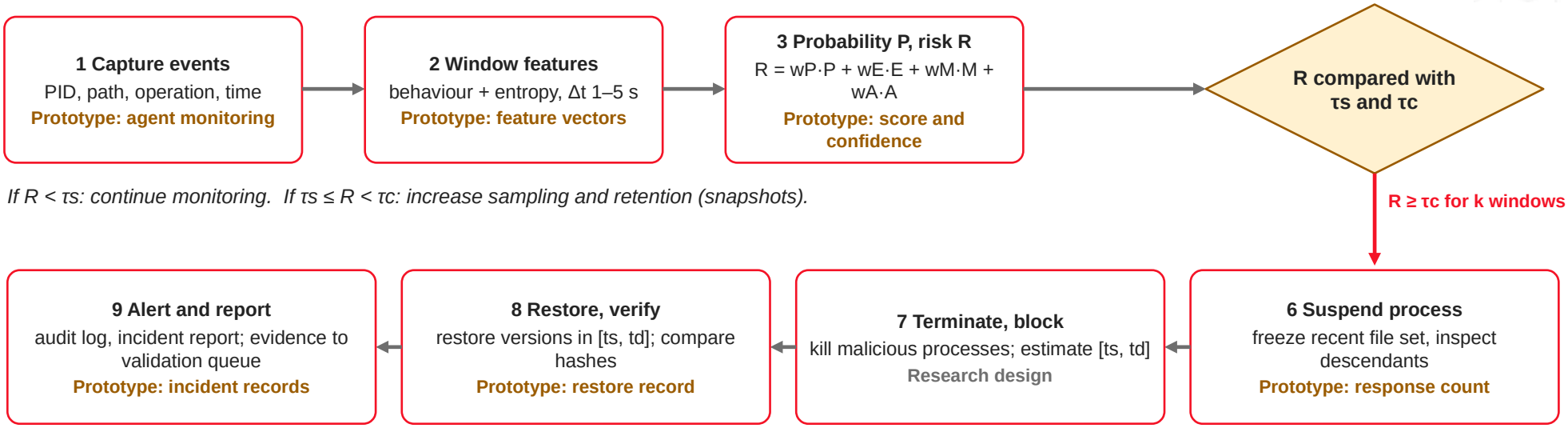
- **Paper's proposed prerequisites:** Windows 10/11 (64-bit), Python 3.10+, 8 GB RAM minimum (16 GB recommended), SQLite or MySQL, scikit-learn-compatible inference, a protected version store and a lightweight dashboard.
- **Actual test machine:** Windows 11 build 10.0.26200 (Device Fleet page). Hardware figures and package versions are recorded with the Stage-II test log.
- **Prototype stack (team-documented):** desktop agent with local SQLite and sync queue, Python + FastAPI backend, Supabase PostgreSQL with RLS, Next.js 16.2.9 dashboard.
- **Model:** the paper compares logistic regression, decision tree, random forest and SVM; the prototype's inference component is compared with this set in Stage-II.
- **Data:** controlled dataset construction follows the paper's §IV-B and is a Stage-II work package.



Hardware & Software Design – Table

Component	Paper prerequisite	Prototype environment	Basis
Endpoint host	Windows 10/11 64-bit; 8 GB RAM min, 16 GB recommended	Windows 11 build 10.0.26200	**Device Fleet page**
Runtime	Python 3.10+	Python 3.12	**Team environment**
Desktop agent	Python agent: monitor, feature service, scikit-learn-style inference, version store	“RansomShield AI Enterprise” v2.0 interface	**Observed**
Persistence	SQLite or MySQL	SQLite local store and sync queue	**Design component**
Backend	Not specified	FastAPI; /health returns HTTP 200	**Observed**
Cloud database	Not specified	Supabase / PostgreSQL with row-level security	**Design component**
Dashboard	Lightweight alerts and reports dashboard	Next.js 16.2.9 web dashboard	**Observed**
Test data	Isolated VM; scenario telemetry + file-entropy records; 22 file formats	Scripted RansomwareTest folder run; controlled dataset follows the paper's design	**Stage-II work package**

Algorithm / Flowchart



Risk score (paper eq. 4)

$R_{j,t} = w_p \cdot P_{j,t} + w_e \cdot E_{j,t} + w_m \cdot M_{j,t} + w_a \cdot A_{j,t}$

Weights are non-negative and sum to one, tuned on validation data; critical when R ≥ τc for k consecutive windows. Proposed research model: the paper does not specify the weights, τs, τc or k.

Entropy features (eqs. 1–3)

$H_S(X) = -\Sigma\, p_i \log_2 p_i$

$H_\alpha(X) = 1/(1-\alpha) \cdot \log_2 \Sigma\, p_i\alpha,\ \ \alpha > 0, \alpha \neq 1$

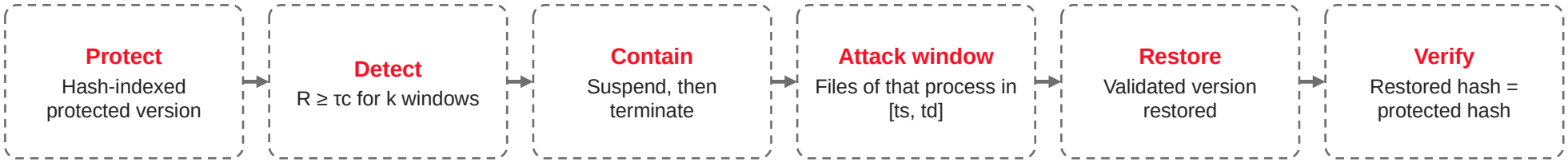
$C_r = S_{original} / S_{compressed}$

Proposed features. The prototype interface displays an entropy-change feature.

Prototype observation

One scripted test incident reached CRITICAL at behaviour score 94/100 (displayed confidence 88–95 %), with ransom-note and extension evidence. This is a displayed score, not a measured accuracy.

Selective Rollback Design



Dashed = research design (paper §III-D)

Proposed (paper §III-D)

Rollback is limited to files tied to the suspected process in [ts, td]. For each file: confirm a protected version exists, verify its stored hash, restore, recompute the hash. Success only if the restored hash matches the protected reference; files without a valid version are reported separately.

Research basis

Prototype observation

Desktop Recovery Center lists one restore: test_target.txt, SUCCESS, 11.6 ms, 19 Jul 2026. Version ID, hash comparison and attack window are not shown, and the CRITICAL web incident shows Recoveries = 0. One logged restore does not by itself establish selective rollback.

Observed in prototype

Verification criteria

RSR = Nrestored / Naffected with valid snapshot × 100 %

Lr = tr − td

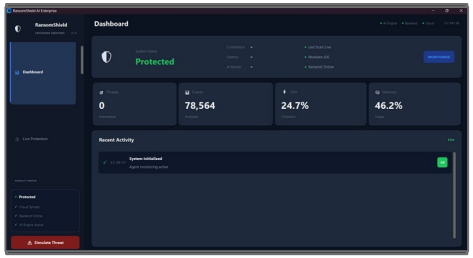
A recovery is verified only when a valid protected version exists, restoration completes and the restored hash matches the protected reference. Measured in Stage-II on synthetic files in an isolated VM.

Stage-II work package

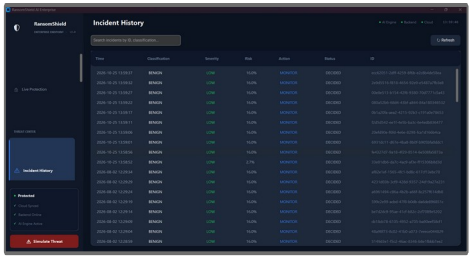
Implementation / Development Status

Module	Paper stage	What the prototype screenshots show	Status
Endpoint monitoring	1–2	Agent: Protected, MONITORING, Modules 6/6, 78,564 events analysed (slide 26)	**Observed**
Feature extraction	3	AI Forensics feature vectors and radar chart (slide 24)	**Observed**
AI inference & decision	4	Benign decisions at 16.0 % risk; one CRITICAL test incident, score 94; AI Forensics label differs from its narrative	**Observed with limitations**
Response / quarantine	5	Incident shows Responses = 1 and status RESPONDING; the action type is not displayed	**Observed with limitations**
Recovery	6–7	One logged restore (test_target.txt, SUCCESS, 11.6 ms); no hash comparison displayed; incident shows Recoveries = 0	**Observed with limitations**
Local store & sync queue	–	SQLite store and upload queue form the endpoint-to-cloud path of the design	**Design component**
FastAPI backend	–	GET /health returns HTTP 200 in the local log (slide 25)	**Observed**
Supabase, enrollment, RLS	–	Enrollment tokens, device ownership and row-level security form the cloud access design	**Design component**
Next.js dashboard	8	Command Center, incident detail, AI Forensics, Device Fleet; summary values on Command Center are inconsistent (slide 27)	**Observed with limitations**
Device fleet & sync	–	Two devices listed with sync times of 26 Jul and 2 Aug 2026	**Observed with limitations**
Audit trail, reports, PDF	8	Navigation entries for audit, reports and PDF export	**Design component**

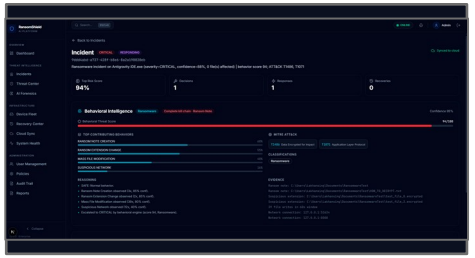
Implementation Evidence – Screenshots



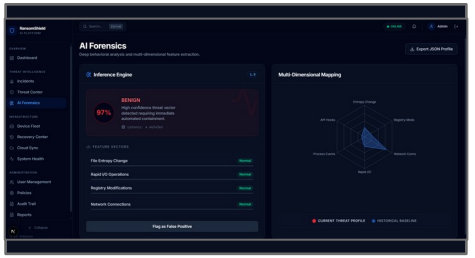
Desktop agent running
Observed – Protected / MONITORING
Detail: slide 26



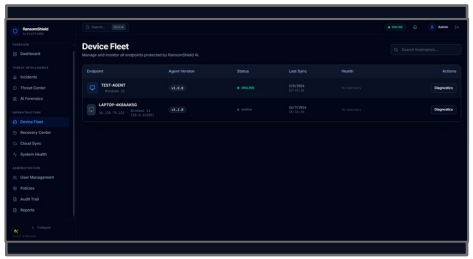
Desktop incident history
Observed – benign decisions only
Detail: slide 26



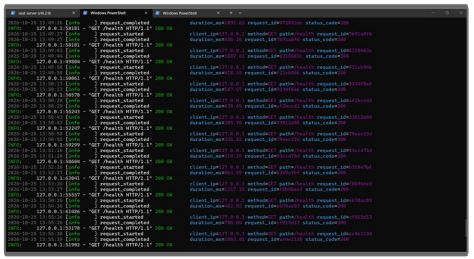
Incident / threat decision
Observed – CRITICAL test incident
Detail: slide 24



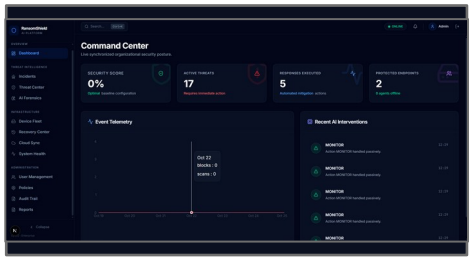
AI forensic features
Observed – UI inconsistency to fix
Detail: slide 24



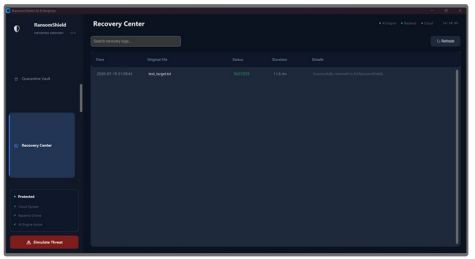
Device fleet & sync
Observed – stale sync times
Detail: slide 25



Backend health log
Observed – /health 200 only
Detail: slide 25



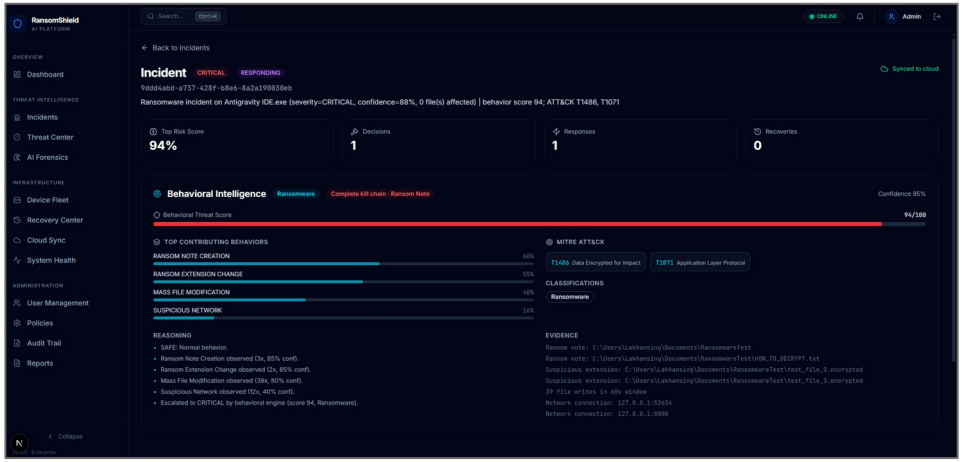
Web dashboard overview
Observed – summary values differ
Detail: slide 27



Recovery record
Observed – one logged restore
Detail: slide 27

Genuine captures supplied by the team; not edited. Thumbnails are small – full-size detail and caveats are on slides 24–27.

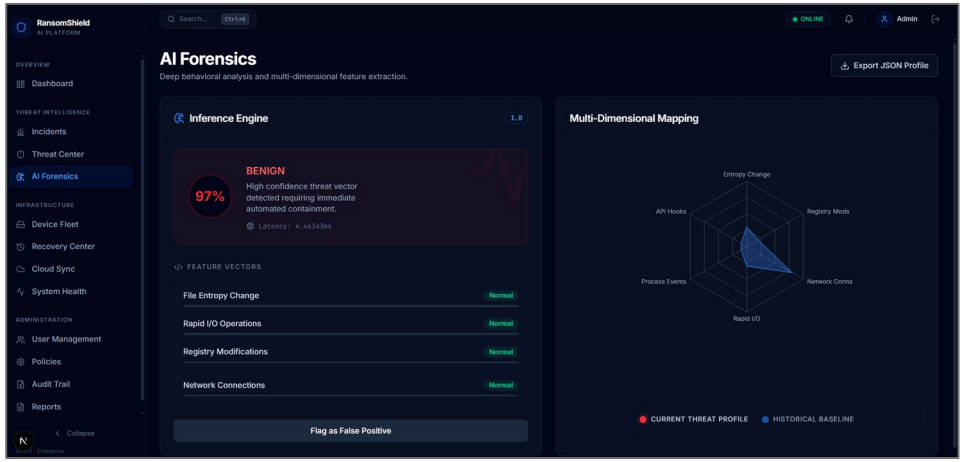
Evidence – Incident & AI Forensics



(a) Incident detail – web dashboard

Shows: CRITICAL / RESPONDING test incident; behaviour score 94/100; confidence 88 % (summary) and 95 % (panel); ATT&CK T1486, T1071; ransom-note files, ".encrypted" extensions and 39 writes in 60 s in a RansomwareTest folder; Decisions 1, Responses 1.

Does not show / caution: Recoveries = 0, 0 files affected. The flagged process is an IDE (Antigravity IDE.exe), probably the launcher of the test script, so process attribution is reviewed in Stage-II. One scripted run is not a detection rate.

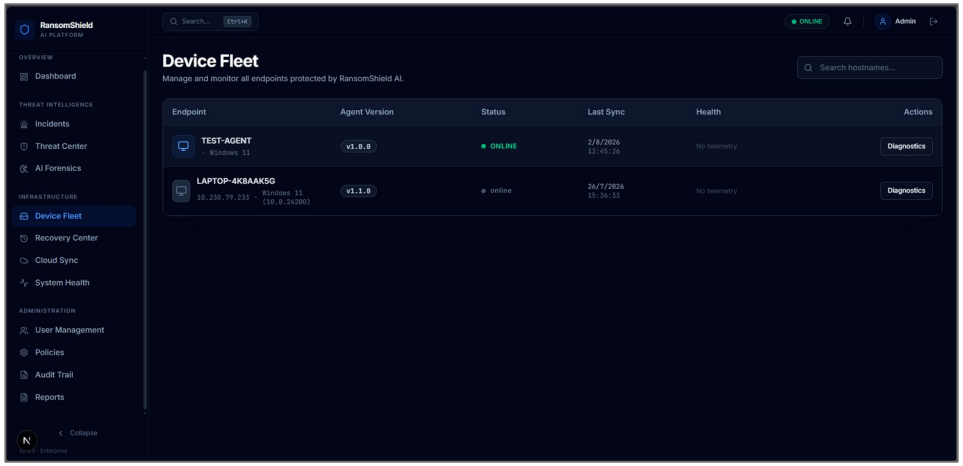


(b) AI Forensics – web dashboard

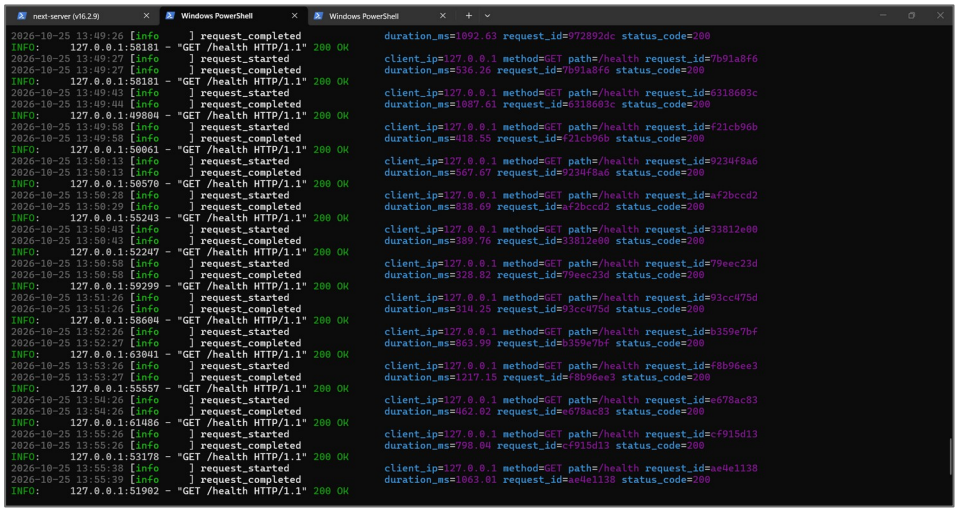
Shows: Inference Engine v1.0 on one record: 97 %, latency 4.46 ms, four feature rows, radar chart, "Flag as False Positive" and export buttons.

Does not show / caution: Headline reads BENIGN while the text reads "High confidence threat vector … containment" and all features read Normal; reconciled in Stage-II work package 1. 4.46 ms is one displayed inference time, not a benchmark.

Evidence – Device Fleet & Backend Health



(a) Device Fleet – web dashboard



(b) Backend /health log – terminal

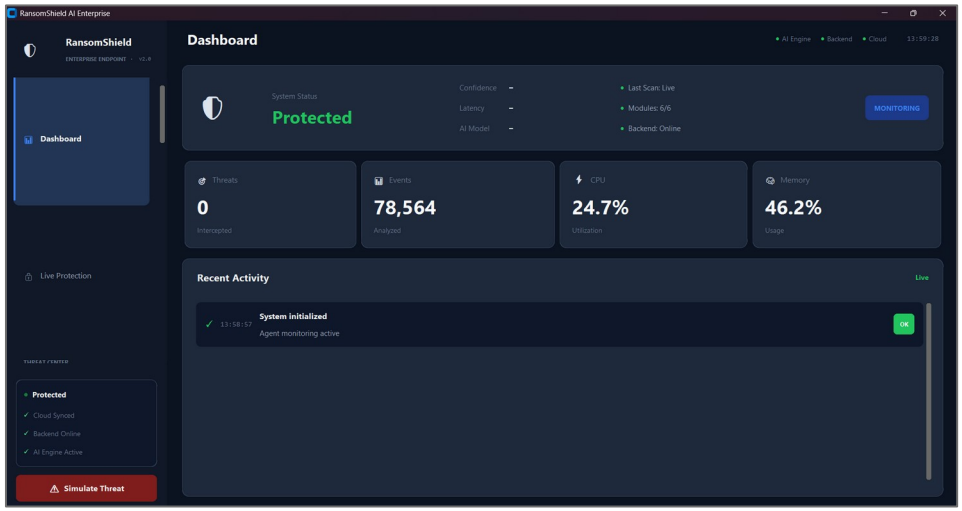
Shows: Two Windows 11 endpoints: TEST-AGENT (v1.0.0) and LAPTOP-4K8AAK5G (v1.1.0, build 10.0.26200); Health column “No telemetry”.

Does not show / caution: Last syncs are 2 Aug and 26 Jul 2026 on a 25 Oct capture while status reads online. One entry is a test agent, so this does not demonstrate multi-device sync. Version differs from the desktop UI (v2.0).

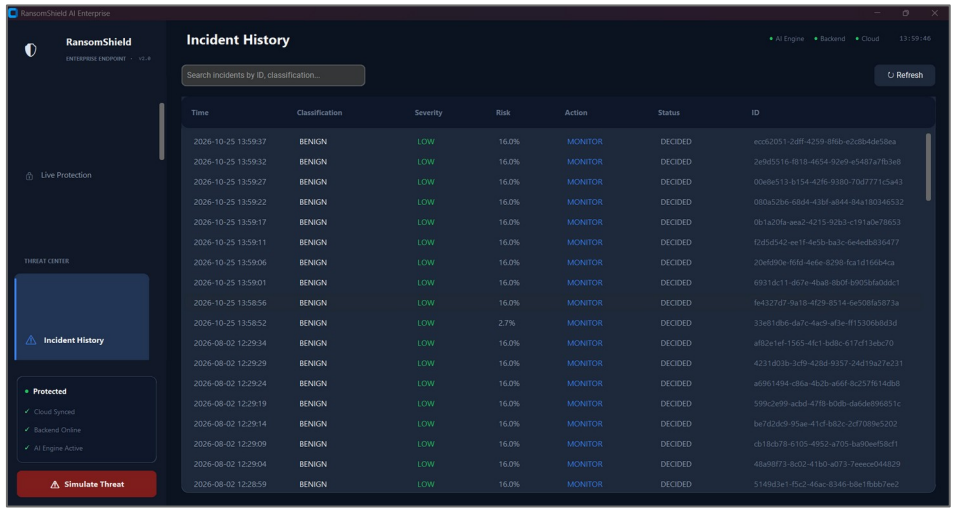
Shows: 25 Oct 2026, 13:49–13:55: GET /health → 200 OK about every 15 s; durations 314–1217 ms across the 14 requests visible; tab “next-server (v16.2.9)”.

Does not show / caution: Proves only that the health route responds. Durations come from an uncontrolled local log, not a benchmark. No authentication, ingestion or RLS behaviour is shown.

Evidence – Desktop Agent



(a) Agent dashboard



(b) Agent incident history

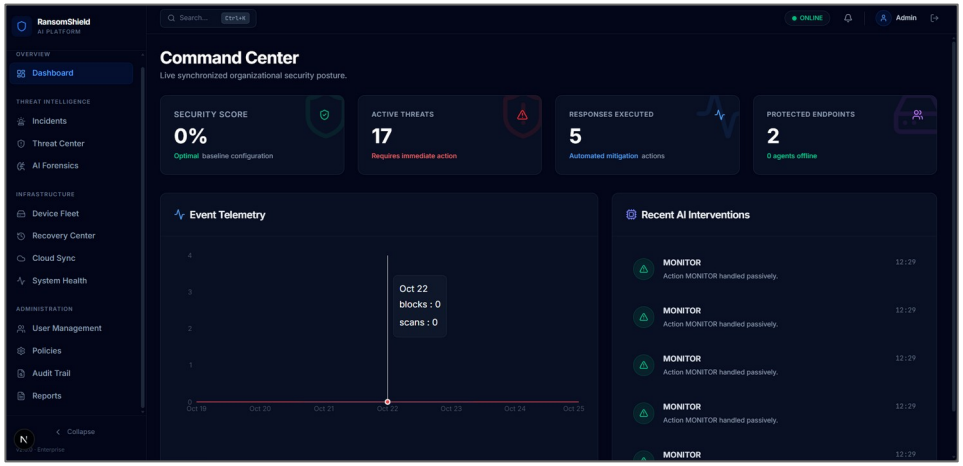
Shows: "RansomShield AI Enterprise" v2.0: Protected, MONITORING, Modules 6/6, Backend Online, 0 threats intercepted, 78,564 events analysed, CPU 24.7 %, memory 46.2 %, a "Simulate Threat" button.

Does not show / caution: Confidence, Latency and AI Model read "–". CPU / memory source (system or agent) unknown, so not an overhead measurement. Sidebar shows empty highlighted areas.

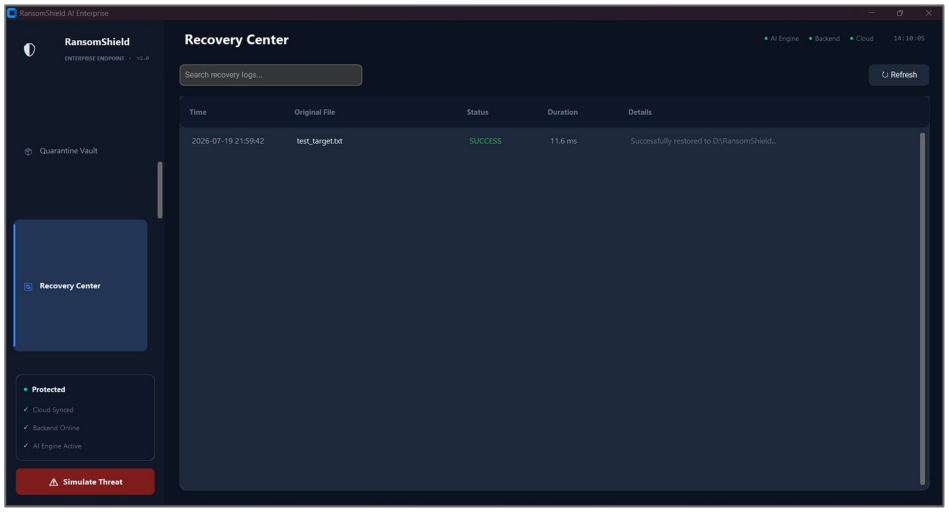
Shows: A decision about every 5 s on 25 Oct 2026 from 13:58:52, plus older entries from 2 Aug 2026: BENIGN, LOW, MONITOR, DECIDED.

Does not show / caution: Risk is 16.0 % in almost every row – looks like a baseline value, so these rows do not show how well benign and malicious cases are separated. No threat, quarantine or recovery appears.

Evidence – Web Overview & Recovery



(a) Command Center – web dashboard



(b) Recovery Center – desktop agent

Shows: Security Score 0 % (“Optimal baseline configuration”), Active Threats 17, Responses Executed 5, Protected Endpoints 2 (0 offline), flat Event Telemetry (Oct 22: blocks 0, scans 0), Recent AI Interventions: MONITOR at 12:29.

Does not show / caution: Summary values differ: 0 % score beside “Optimal”; 17 active threats beside flat telemetry and a benign-only agent log; 2 endpoints protected beside old sync times. These are displayed values, not validated metrics.

Shows: One recovery entry: test_target.txt, SUCCESS, 11.6 ms, “Successfully restored to D:\RansomShield…”, dated 19 Jul 2026 21:59:42; a Quarantine Vault menu item.

Does not show / caution: Single file, three months before other captures. No protected-version ID, hash comparison, attack window or link to the CRITICAL incident (Recoveries = 0). A logged success does not validate selective rollback.

Preliminary Results / Validation

Stage-I prototype demonstration. Endpoint monitoring interface, incident decision display, dashboard views, one separately logged restoration record and a responding backend health endpoint. The supplied evidence does not establish model accuracy, end-to-end containment effectiveness or hash-verified recovery.

Observed evidence	What it demonstrates	What it does not establish
Scripted test incident (slide 24)	The prototype raised a CRITICAL incident with a behaviour score of 94/100 and ransom-note, extension-change and mass-write evidence	A displayed score is not classification accuracy; one scripted run gives no detection rate or false-positive rate
Agent decision log (slide 26)	Continuous decisions classified BENIGN / LOW with action MONITOR	Separation of benign and malicious activity; the risk value is almost constant at 16.0 %
Recovery record (slide 27)	A restore of test_target.txt is logged as SUCCESS (11.6 ms) and is shown separately from the CRITICAL incident	A hash comparison against a protected version, or a link to that incident
Backend health log (slide 25)	The /health route returns HTTP 200 on the local server	Authenticated ingestion, row-level security or end-to-end synchronisation
Dashboard views (slides 24–27)	Command Center, incident detail, AI Forensics and Device Fleet render from the prototype	Consistent summary values; some displayed figures differ between views

Performance Evaluation Plan

- **Protocol (paper §IV):** stratified 70:30 train-test split; ten-fold cross-validation on the training partition; windows from the same executable never cross partitions; family hold-out evaluation where feasible.
- **Ablation:** behaviour-only, entropy-only and hybrid features; logistic regression, decision tree, random forest and SVM. The hybrid is judged beneficial only if it improves early detection or reduces false alarms at acceptable overhead.
- **Metrics:** precision, recall, F1, PR-AUC; detection latency; files modified before suspension; false alarms per hour; rollback success; recovery time; CPU, memory and disk overhead; sync reliability.
- **Scope:** this protocol defines how Stage-II results are produced. Stage-I reports the observed prototype baseline on slide 28 and makes no comparison with commercial products.

Safe-testing controls

- Authorised isolated VM, shared folders disabled, limited networking, disposable snapshots.
- Synthetic files only; scripted, reversible ransomware-like activity.
- No live ransomware, payloads or credential theft.
- Benign heavy-I/O workloads: backup, compression, sync, installers, bulk rename.
- Hash manifest before and after each run.

Performance Evaluation – Table

Metric	Definition and procedure	Evidence	Acceptance rule	Observed baseline
Precision, recall, F1, PR-AUC	TP/(TP+FP), TP/(TP+FN), eqs. (6)–(7); held-out windows, family hold-out	Confusion matrix, PR curve	Hybrid improves on both single-feature configurations; targets fixed before testing	No classification metric is reported at Stage-I
False alarms per hour	Alarms on benign heavy-I/O workloads per hour of runtime	Alarm log	Fewer alarms than the behaviour-only baseline	Agent log shows benign decisions only; no hourly rate
Detection latency Ld	Ld = td − ts, in seconds, from the event log	Timestamps	Reported as mean and median per scenario	4.46 ms displayed for one inference (not Ld)
Files modified before suspension	Files changed between ts and suspension	File-change log	Lower is better across configurations	Test incident shows 0 files affected
Rollback success RSR	Nrestored / Naffected with valid snapshot × 100 %, eq. (5)	Hash manifest	Counted only when restored hash equals the protected reference	One restore logged; no hash comparison displayed
Recovery time Lr	Lr = tr − td, in seconds, to verified restore	Timestamps	Reported as mean and median	11.6 ms displayed for one restored file
CPU, memory, disk overhead	Resource use idle and under load, monitor on vs off	Performance counters	Overhead reported against the monitor-off baseline	UI shows 24.7 % CPU, 46.2 % memory (source not stated)
Sync reliability, API response	Events delivered after offline periods; request time in ms	Queue and API logs	Every queued event delivered after reconnection	/health returns 200; 314–1217 ms in an informal log

Industry / Domain Expert Methodology Validation

Scope at Stage-I. No external expert validation is claimed in this presentation. The methodology follows the research manuscript and is checked here against the criteria below.

Review criterion	What a reviewer examines	Evidence used
Feature model	Behaviour and entropy features are justified for early detection and for benign heavy-I/O workloads	Paper §II–III; slide 20
Decision logic	Staged thresholds and consecutive-window confirmation limit false alarms	Paper §III-B/C; eq. (4)
Recovery claim	A restore counts only with a valid protected version and a matching hash	Paper §III-D; slide 21
Test safety	Isolated VM, synthetic files, no live malware	Paper §IV-A; slide 29
Security design	Enrollment, device ownership and row-level security limit access to each owner's data	Slides 16–17

Innovation, Novelty & Industry Relevance

Proposed contributions (paper)

Research basis

- C1 hybrid behaviour and file-entropy analysis
- C2 short time-window feature extraction
- C3 confidence-based process-level risk engine
- C4 attack-window-aware selective rollback
- C5 operational evaluation of detection, damage, recovery and overhead

How they fit together

Design component

- Windows of behaviour and entropy features give an early risk score per process.
- Staged thresholds decide between monitoring, suspension and termination.
- The attack window bounds which files are restored, and hashes verify the result.

Observed in the prototype

Observed in prototype

- Agent, dashboard and backend health endpoint running.
- A scripted test reached CRITICAL at score 94/100.
- One restore logged. No claim of live-ransomware prevention.

Relevance and status

Stage-II work package

- Relevant to Windows endpoints and small organisations; the paper links the work to SDG 9, 16 and 8.
- Compared with [9], [10] and [28], the proposal differs in combining hybrid features, staged interception and attack-window rollback; superiority is not claimed.
- An acceptance notification for the manuscript (Paper ID 8) was received on 8 October 2026; publication follows the camera-ready requirements.

Individual Contribution

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Lakhansing Narsing Patil

Seat no. 112023454

Backend, cloud synchronisation and AI detection

Contribution area: the FastAPI backend, cloud synchronisation with Supabase, and the AI detection component of the prototype.

Rutuja Pankaj Patil

Seat no. 112023492

Documentation and technical reporting

Contribution area: project documentation and technical reporting, including the written description of the system and the research work.

Srushti Vinayak Khapre

Seat no. 112023335

Testing and test execution

Contribution area: preparing and running tests on the prototype and recording their outcomes.

Vaishnavi Sunil Thapekar

Seat no. 112023490

Backend checks, integration checks and testing

Contribution area: checking the backend and its integration with the other components, and supporting testing.

Individual Contribution – Table

Student	Contribution area	Scope
Lakhansing Narsing Patil [112023454]	Backend, cloud synchronisation and AI detection	The FastAPI backend, cloud synchronisation with Supabase, and the AI detection component of the prototype.
Rutuja Pankaj Patil [112023492]	Documentation and technical reporting	Project documentation and technical reporting, including the written description of the system and the research work.
Srushti Vinayak Khapre [112023335]	Testing and test execution	Preparing and running tests on the prototype and recording their outcomes.
Vaishnavi Sunil Thapekar [112023490]	Backend checks, integration checks and testing	Checking the backend and its integration with the other components, and supporting testing.

Responsibilities follow the allocation specified by the project team; the whole team contributed to the integrated prototype and this presentation.

Challenges & Mitigation

Challenge	Observation in the project	Mitigation / verification method
Legitimate heavy I/O vs malicious activity	Behavioural overlap [4]–[6]; high entropy also occurs in archives [7]	Benign workloads in the dataset; consecutive-window confirmation; hybrid features
Process attribution	The scripted incident names an IDE process; confidence reads 88 % and 95 % in different parts of the display	Check process lineage logging; draw every figure from one decision record
AI Forensics and Command Center consistency	BENIGN label beside a threat narrative; 0 % score beside “Optimal”; 17 active threats beside flat telemetry	Trace each displayed field to its source value and align the display with the decision record
Device sync and version display	Fleet shows old sync times, no telemetry and versions v1.0.0 / v1.1.0 against the UI's v2.0	Verify heartbeat updates and the version string reported at enrollment
Recovery evidence	A restore is logged without a displayed hash comparison or incident link	Store original and restored hashes and link each restore to its incident
Controlled datasets and quantitative validation	Stage-I holds prototype observations and no quantified performance	Build the dataset per paper §IV-B and run the Stage-II protocol

Stage-II Work Plan & Expected Outcomes

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WP1
Reconcile inconsistencies

Incident, process-attribution and dashboard values aligned to one decision record.

WP2
Capture clear evidence

Consistent screenshots and logs from the running system.

WP3
Dataset and safe scenarios

Documented dataset construction and isolated-VM test scenarios.

WP4
Feature configurations

Behaviour-only, entropy-only and hybrid results.

WP5
Model comparison

LR, DT, RF and SVM evaluated where implemented.

WP6
Measure the pipeline

Detection latency, files changed, false alarms, recovery success and time, overhead.

WP7
Security and sync tests

Device ownership, authentication, cloud sync and retry behaviour.

WP8
Manuscript and demo

Camera-ready paper and final demonstration.

Work packages run in sequence; each ends with a documented deliverable.

Conclusion

Signature-based tools may not react in time to new ransomware variants, and behavioural detectors must still tell malicious encryption from legitimate heavy file work. This project therefore studies a hybrid framework that combines short time-window behavioural features with file-entropy analysis, a confidence-based risk engine for staged interception, and attack-window selective rollback verified by hashes.

The RansomShield AI prototype brings these ideas together as a desktop endpoint agent, a FastAPI backend with Supabase storage and a Next.js dashboard. The Stage-I evidence shows the monitoring interface, a CRITICAL scripted incident with a behaviour score of 94/100, dashboard views, one logged restoration and a responding health endpoint. This evidence is preliminary: the displayed score is not an accuracy figure, and measured detection performance and hash-verified recovery require controlled validation.

Stage-II therefore reconciles the observed inconsistencies, builds the dataset in an isolated environment, compares feature configurations and models, and measures latency, damage, recovery and overhead, leading to the camera-ready manuscript and final demonstration.

References

Research publications

[4] H. Oz, A. Aris, A. Levi, A. S. Uluagac, “A survey on ransomware: Evolution, taxonomy, and defense solutions,” arXiv:2102.06249, 2021.

[5] A. Vehabovic et al., “Ransomware detection and classification strategies,” arXiv:2304.04398, 2023.

[6] M. Ozer et al., “A prevention and detection system for ransomware attacks,” arXiv:2001.02282, 2020.

[7] C.-M. Hsu et al., “Enhancing file entropy analysis to improve machine learning detection rate of ransomware,” IEEE Access, vol. 9, pp. 138345–138357, 2021, doi: 10.1109/ACCESS.2021.3114148.

[8] D. Smith, S. Khorsandroo, K. Roy, “Machine learning algorithms and frameworks in ransomware detection,” IEEE Access, vol. 10, pp. 117597–117613, 2022, doi: 10.1109/ACCESS.2022.3218779.

[9] N. Scaife, H. Carter, P. Traynor, K. R. B. Butler, “CryptoLock (and Drop It): Stopping ransomware attacks on user data,” in Proc. IEEE ICDCS, 2016, pp. 303–312. †

[10] A. Continella et al., “ShieldFS: A self-healing, ransomware-aware filesystem,” in Proc. ACM AsiaCCS, 2016, pp. 336–347 (as printed in the manuscript). †

Research publications (continued)

[25] K. Cabaj, W. Mazurczyk, “Using software-defined networking for ransomware mitigation: The case of CryptoWall,” IEEE Network, vol. 30, no. 6, pp. 14–20, 2016.

[26] A. Kharraz et al., “Cutting the Gordian knot: A look under the hood of ransomware attacks,” in Proc. DIMVA, 2015, pp. 3–24.

[27] D. Sgandurra et al., “Automated dynamic analysis of ransomware: Benefits, limitations and use for detection,” arXiv:1609.03020, 2016.

[28] S. Mehnaz, A. Mudgerikar, E. Bertino, “RWGuard: A real-time detection system against cryptographic ransomware,” in Proc. RAID, 2018, pp. 114–136.

Standards, guidance and documentation

[1] J. Cawthra et al., Data Integrity: Identifying and Protecting Assets Against Ransomware and Other Destructive Events, NIST SP 1800-25, 2020.

[2] J. Cawthra et al., Data Integrity: Detecting and Responding to Ransomware and Other Destructive Events, NIST SP 1800-26, 2020.

[3] T. McBride et al., Data Integrity: Recovering from Ransomware and Other Destructive Events, NIST SP 1800-11, 2020.

[S1] Supabase, “Row Level Security,” Supabase Documentation. [S2] FastAPI Documentation. [S3] Next.js Documentation.

Numbering follows the manuscript's bibliography of 30 entries, of which 11 are research publications. † Details are as printed in the manuscript, with [9] author initial corrected to N. Scaife; to be rechecked against publisher records during camera-ready preparation.

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THANK YOU

Questions & Discussion